

Further Probability & Statistics

University-Preparation Notes
Chapters 8–12

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Level: A-Level Further Mathematics
Audience: Aspiring Cambridge, Oxford, Imperial,
Warwick, MIT, Stanford applicants

These notes build deep mathematical understanding of Further Statistics. Topics include continuous random variables, inferential statistics, chi-squared tests, non-parametric tests, and probability generating functions.

Preface

These notes are written for the ambitious Further Mathematics student whose sights are set beyond merely passing examinations. The goal is to build *genuine statistical thinking*: the ability to reason carefully about uncertainty, to choose appropriate methods, to understand what those methods actually do, and to interpret results correctly.

Statistics is not a collection of procedures to be memorised. It is a principled framework for learning from data in the presence of randomness. Every formula here has a story behind it—a problem someone needed to solve, an insight someone had, and a derivation that makes the formula inevitable rather than arbitrary.

How to use these notes. Read each section in order the first time. When you encounter a result, try to anticipate the derivation before reading it. Work every example yourself before looking at the solution. The exploration and extension questions at the end of each chapter are designed to take you beyond the syllabus into the territory you will encounter in your first year of university.

Prerequisites. These notes assume familiarity with A-Level Mathematics (probability, statistics, calculus) and the earlier chapters of Further Statistics (discrete distributions, sampling, hypothesis testing with the normal and Poisson distributions, linear regression and correlation).

Contents

Chapter 1

Continuous Random Variables

1.1 Introduction

So far you have studied *discrete* random variables: quantities that take values in a finite or countably infinite set (such as the number of heads in ten coin flips, or the number of calls arriving at a switchboard in one minute). For discrete variables, we can assign a probability to each individual outcome.

But many phenomena in the real world are *inherently continuous*: the exact height of a randomly selected adult, the time until a radioactive atom decays, the return on a financial investment over a year. For such quantities, the outcome can be *any* real number in some interval, and it becomes meaningless to ask for the probability of any *exact* value.

Intuition

If you pick a real number uniformly at random from $[0, 1]$, what is the probability that you pick exactly 0.5? The answer is zero—not because it is impossible, but because the interval contains infinitely many points and no single point has positive measure. What *does* make sense is asking for the probability of landing in an interval, e.g., $\mathbb{P}(0.3 \leq X \leq 0.7) = 0.4$.

Continuous random variables arise throughout science and engineering:

- **Physics:** lifetimes of particles, measurement errors.
- **Finance:** stock prices, option payoffs, interest rates.
- **Biology:** heights, weights, enzyme concentrations.
- **Engineering:** stress tolerances, signal noise, waiting times.
- **Machine learning:** model parameters treated as continuous latent variables.

1.2 The Probability Density Function

1.2.1 Definition and Intuition

Definition 1.2.1 (Probability Density Function). A random variable X is said to be *continuous* if there exists a non-negative function $f : \mathbb{R} \rightarrow [0, \infty)$, called the **probability**

density function (PDF) of X , such that for all $a \leq b$,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) \, dx.$$

The PDF must satisfy two conditions:

(i) **Non-negativity:** $f(x) \geq 0$ for all $x \in \mathbb{R}$.

(ii) **Normalisation:** $\int_{-\infty}^{\infty} f(x) \, dx = 1$.

Intuition

Think of the PDF as a *density of probability per unit length*. A height $f(x)$ does *not* represent a probability; indeed $f(x)$ can be greater than 1. The *area* under the curve between two points represents the probability of the variable falling between those points. This is exactly the way a histogram works: the area of each bar gives the relative frequency (proportion) of observations in that interval.

Probability as area under the PDF

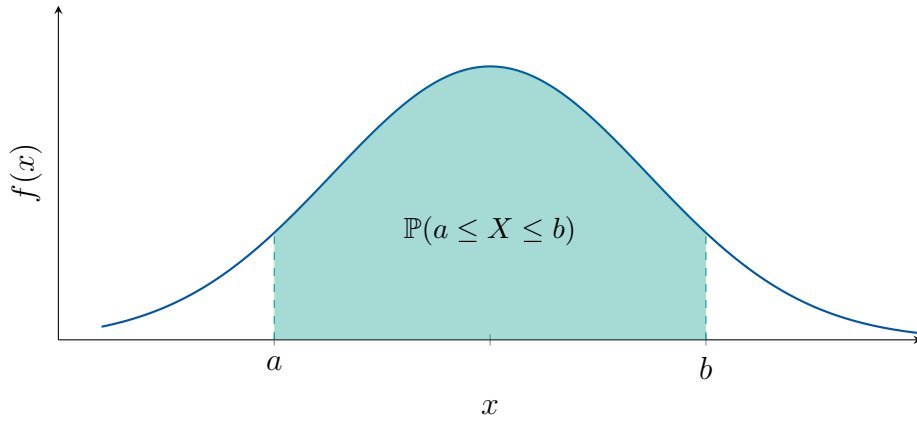


Figure 1.1: The probability $\mathbb{P}(a \leq X \leq b)$ equals the shaded area under the PDF between $x = a$ and $x = b$.

1.2.2 Key Properties

Because probabilities correspond to areas rather than heights, several important properties follow immediately:

Proposition 1.2.1. *For a continuous random variable X with PDF f :*

(i) $\mathbb{P}(X = a) = 0$ for every $a \in \mathbb{R}$.

(ii) $\mathbb{P}(a < X < b) = \mathbb{P}(a \leq X < b) = \mathbb{P}(a < X \leq b) = \mathbb{P}(a \leq X \leq b)$.

(iii) $\mathbb{P}(X \geq a) = \int_a^{\infty} f(x) \, dx$.

(iv) $\mathbb{P}(X \leq a) + \mathbb{P}(X > a) = 1$.

Property (i) is crucial: the probability of any *specific* value is zero. This does *not* mean that value cannot occur—it simply reflects the nature of the continuous setting.

Exam Trap

Many students write $\mathbb{P}(X = a) = f(a)$. This is **wrong**. $f(a)$ is a *density*, not a probability. The probability of any single point is zero. Only integrals (areas) give probabilities.

1.2.3 Finding Probabilities: Worked Examples

Worked Example 1.2.1 — Verifying a PDF and Computing Probabilities

Let X have PDF

$$f(x) = \begin{cases} kx(4-x) & 0 \leq x \leq 4 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the value of k .
- (b) Find $\mathbb{P}(1 \leq X \leq 3)$.
- (c) Find $\mathbb{P}(X > 2.5)$.

Solution. (a) The normalisation condition requires $\int_{-\infty}^{\infty} f(x) dx = 1$. Since $f(x) = 0$ outside $[0, 4]$:

$$\int_0^4 kx(4-x) dx = k \int_0^4 (4x - x^2) dx = k \left[2x^2 - \frac{x^3}{3} \right]_0^4 = k \left(32 - \frac{64}{3} \right) = k \cdot \frac{32}{3}.$$

Setting this equal to 1: $k = \frac{3}{32}$.

(b)

$$\mathbb{P}(1 \leq X \leq 3) = \frac{3}{32} \int_1^3 x(4-x) dx = \frac{3}{32} \left[2x^2 - \frac{x^3}{3} \right]_1^3 = \frac{3}{32} \left[(18 - 9) - \left(2 - \frac{1}{3} \right) \right] = \frac{3}{32} \cdot \frac{20}{3} = \frac{20}{32} = \frac{5}{8}.$$

(c)

$$\mathbb{P}(X > 2.5) = \frac{3}{32} \int_{2.5}^4 x(4-x) dx = \frac{3}{32} \left[2x^2 - \frac{x^3}{3} \right]_{2.5}^4.$$

At $x = 4$: $2(16) - \frac{64}{3} = 32 - \frac{64}{3} = \frac{32}{3}$. At $x = 2.5$: $2(6.25) - \frac{15.625}{3} = 12.5 - \frac{15.625}{3} = \frac{22.375}{3}$.

$$\mathbb{P}(X > 2.5) = \frac{3}{32} \cdot \frac{32 - 22.375}{3} = \frac{9.625}{32} = \frac{77}{256} \approx 0.301.$$

□

1.2.4 Common Continuous Distributions

Definition 1.2.2 (Uniform Distribution). $X \sim U(a, b)$ if

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b.$$

Every sub-interval of equal length has the same probability.

Definition 1.2.3 (Exponential Distribution). $X \sim \text{Exp}(\lambda)$ if

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0, \quad \lambda > 0.$$

It models the time until the first event in a Poisson process with rate λ . It has the remarkable *memoryless property*: $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$.

Exponential PDFs for different rates λ

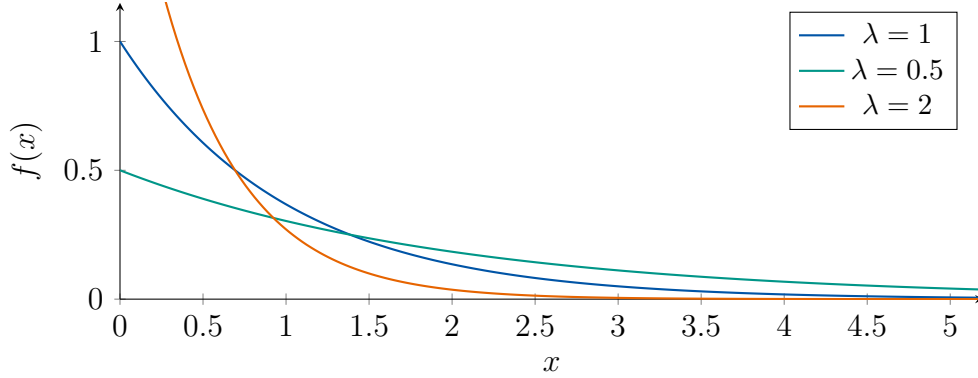


Figure 1.2: Exponential PDFs: larger λ means faster decay and shorter expected waiting times.

1.3 The Cumulative Distribution Function

1.3.1 Definition

Definition 1.3.1 (Cumulative Distribution Function). The **cumulative distribution function** (CDF) of a random variable X is defined for all $x \in \mathbb{R}$ by

$$F(x) = \mathbb{P}(X \leq x).$$

For a continuous random variable,

$$F(x) = \int_{-\infty}^x f(t) dt.$$

The CDF accumulates probability as x increases.

1.3.2 Properties of the CDF

Proposition 1.3.1 (Properties of the CDF). *For any CDF F :*

- (i) $0 \leq F(x) \leq 1$ for all x .
- (ii) F is non-decreasing: if $a < b$ then $F(a) \leq F(b)$.
- (iii) $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow +\infty} F(x) = 1$.
- (iv) For a continuous RV, F is continuous everywhere.
- (v) The PDF is the derivative of the CDF: $f(x) = F'(x)$ wherever the derivative exists.

The fifth property is the Fundamental Theorem of Calculus applied to probability. It tells us that the PDF and CDF are two sides of the same coin: one is the derivative of the other; the other is the integral of the first.

PDF (area interpretation) and its CDF

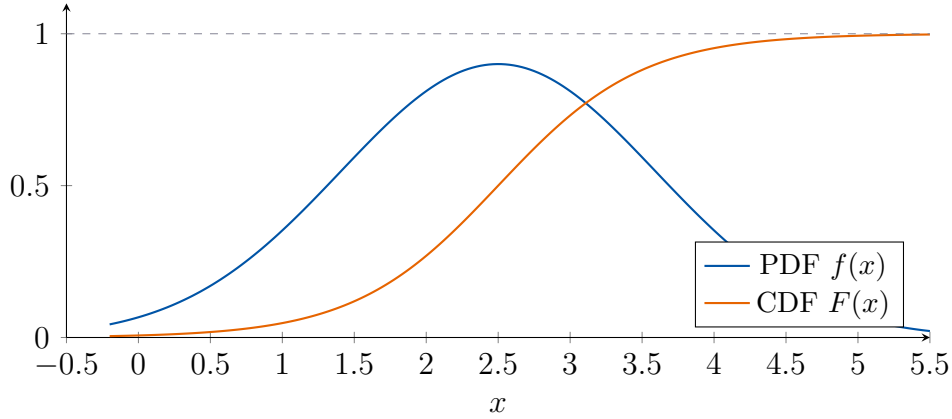


Figure 1.3: The CDF starts at 0, is S-shaped and non-decreasing, and asymptotes to 1. It is the running integral of the PDF.

1.3.3 Using the CDF

Once we have $F(x)$, computing probabilities is straightforward:

$$\mathbb{P}(a \leq X \leq b) = F(b) - F(a), \quad \mathbb{P}(X > a) = 1 - F(a).$$

Worked Example 1.3.1 — Deriving the CDF

For the PDF $f(x) = \frac{3}{32}x(4-x)$, $0 \leq x \leq 4$, find $F(x)$ and hence find the median.

Solution. For $x < 0$: $F(x) = 0$.

For $0 \leq x \leq 4$:

$$F(x) = \int_0^x \frac{3}{32}t(4-t) dt = \frac{3}{32} \left[2t^2 - \frac{t^3}{3} \right]_0^x = \frac{3}{32} \left(2x^2 - \frac{x^3}{3} \right) = \frac{x^2}{16} \left(3 - \frac{x}{4} \right).$$

For $x > 4$: $F(x) = 1$.

Median m : Solve $F(m) = 0.5$:

$$\frac{m^2(12-m)}{64} = 0.5 \implies m^2(12-m) = 32.$$

This is a cubic. The distribution is symmetric about $x = 2$ (check: the PDF $f(x) = f(4-x)$), so the median is $m = 2$. $\square$

1.3.4 Quantiles, Percentiles, and the Median

Definition 1.3.2 (Quantile). The p -th quantile (or $100p$ -th percentile) of a continuous random variable X is the value x_p satisfying $F(x_p) = p$, equivalently $\mathbb{P}(X \leq x_p) = p$.

Special cases:

- $p = 0.5$: the **median** m .
- $p = 0.25$: the **lower quartile** Q_1 .
- $p = 0.75$: the **upper quartile** Q_3 .

For a symmetric unimodal distribution, mean = median = mode.

1.4 Expectation of a Function of a Continuous Random Variable

1.4.1 Expected Value

Definition 1.4.1 (Expected Value). For a continuous RV X with PDF f , the **expected value** (mean) is

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x f(x) dx,$$

provided the integral converges absolutely.

Intuition

$\mathbb{E}(X)$ is the long-run average value of X over many independent repetitions. It is the *centre of mass* of the probability density: if you placed mass $f(x) dx$ at position x along a weightless rod, $\mathbb{E}(X)$ is where you would place a fulcrum to balance it.

1.4.2 The Law of the Unconscious Statistician (LOTUS)

Theorem 1.4.1 (LOTUS for Continuous RVs). If X has PDF f and $g : \mathbb{R} \rightarrow \mathbb{R}$ is a measurable function, then

$$\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x) f(x) dx,$$

provided the integral converges absolutely.

Remark 1.4.1. The name is humorous: students sometimes forget this theorem and compute $\mathbb{E}(g(X))$ by first finding the distribution of $g(X)$ and then integrating. The theorem says you can use the original PDF—a major computational shortcut.

Key applications of LOTUS:

$$\begin{aligned}\mathbb{E}(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx, \\ \text{Var}(X) &= \mathbb{E}(X^2) - [\mathbb{E}(X)]^2, \\ \mathbb{E}(aX + b) &= a\mathbb{E}(X) + b.\end{aligned}$$

Definition 1.4.2 (Variance and Standard Deviation).

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}(X^2) - \mu^2, \quad \text{where } \mu = \mathbb{E}(X).$$

The **standard deviation** is $\sigma = \sqrt{\text{Var}(X)}$.

Worked Example 1.4.1 — Expectation and Variance

For $f(x) = \frac{3}{32}x(4-x)$, $0 \leq x \leq 4$, find $\mathbb{E}(X)$, $\mathbb{E}(X^2)$, $\text{Var}(X)$, and $\mathbb{E}(3X^2 - 2X + 1)$.

Solution.

$$\mathbb{E}(X) = \int_0^4 x \cdot \frac{3}{32}x(4-x) \, dx = \frac{3}{32} \int_0^4 (4x^2 - x^3) \, dx = \frac{3}{32} \left[\frac{4x^3}{3} - \frac{x^4}{4} \right]_0^4 = \frac{3}{32} \left(\frac{256}{3} - 64 \right) = \frac{3}{32} \cdot \frac{64}{3} =$$

$$\mathbb{E}(X^2) = \frac{3}{32} \int_0^4 (4x^3 - x^4) \, dx = \frac{3}{32} \left[x^4 - \frac{x^5}{5} \right]_0^4 = \frac{3}{32} \left(256 - \frac{1024}{5} \right) = \frac{3}{32} \cdot \frac{256}{5} = \frac{24}{5} = 4.8.$$

$$\text{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = 4.8 - 4 = 0.8.$$

By linearity of expectation:

$$\mathbb{E}(3X^2 - 2X + 1) = 3\mathbb{E}(X^2) - 2\mathbb{E}(X) + 1 = 3(4.8) - 2(2) + 1 = 14.4 - 4 + 1 = 11.4.$$

□

1.4.3 Mode and Median of a Continuous Distribution

Definition 1.4.3 (Mode). The **mode** of a continuous distribution is the value of x at which $f(x)$ is maximised. It is found by solving $f'(x) = 0$ and checking it is a maximum.

Note that a distribution may be **bimodal** (two local maxima) or even **multimodal**.

Worked Example 1.4.2 — Finding the Mode

For $f(x) = \frac{3}{32}x(4-x)$, $0 \leq x \leq 4$:

Solution.

$$f'(x) = \frac{3}{32}(4 - 2x) = 0 \implies x = 2.$$

Since $f''(2) = \frac{3}{32}(-2) < 0$, this is a maximum. The mode is $x = 2$.

□

1.5 Finding the PDF and CDF of $Y = g(X)$

1.5.1 Why Transformations Matter

Often we know the distribution of X but we want the distribution of a function $Y = g(X)$. For example:

- If X is the radius of a sphere (uniform), what is the distribution of its volume $Y = \frac{4}{3}\pi X^3$?

- If $X \sim \text{Exp}(\lambda)$, what is the distribution of $Y = X^2$ (which arises in Weibull reliability models)?

1.5.2 The CDF Method

The most general and reliable approach is the **CDF method**:

Step 1: Write $F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y)$.

Step 2: Rearrange the inequality to get bounds on X .

Step 3: Express in terms of F_X .

Step 4: Differentiate to get $f_Y(y) = F'_Y(y)$.

Worked Example 1.5.1 — Monotone Increasing Transformation

Let X have PDF $f_X(x) = 2x$, $0 \leq x \leq 1$. Find the PDF of $Y = X^3$.

Solution. **Step 1.** The support of Y is $[0, 1]$. For $y \in [0, 1]$:

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X^3 \leq y) = \mathbb{P}(X \leq y^{1/3}) = F_X(y^{1/3}) = \int_0^{y^{1/3}} 2t \, dt = y^{2/3}.$$

Step 2. Differentiate:

$$f_Y(y) = \frac{d}{dy} y^{2/3} = \frac{2}{3} y^{-1/3}, \quad 0 < y \leq 1.$$

□

1.5.3 The Change-of-Variables Formula (Jacobian Method)

For a **strictly monotone** transformation $y = g(x)$, there is a slick formula:

Theorem 1.5.1 (Change of Variables — One Dimension). *Let X have PDF f_X . If $Y = g(X)$ where g is strictly monotone and differentiable, then*

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

Proof. **Case: g strictly increasing.**

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

Differentiating: $f_Y(y) = f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$. Since g is increasing, g^{-1} is increasing, so the derivative is positive and equals the absolute value.

Case: g strictly decreasing.

$$F_Y(y) = \mathbb{P}(g(X) \leq y) = \mathbb{P}(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y)).$$

Differentiating: $f_Y(y) = -f_X(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y)$. Since g is decreasing, g^{-1} is decreasing, so $\frac{d}{dy} g^{-1}(y) < 0$ and the product is positive. Taking the absolute value unifies both cases. □

Worked Example 1.5.2 — Applying the Jacobian Formula

Let $X \sim U(0, 1)$. Find the PDF of $Y = -\ln X$.

Solution. $g(x) = -\ln x$ is strictly decreasing on $(0, 1)$. The inverse: $x = g^{-1}(y) = e^{-y}$. The support of Y : as $x \in (0, 1)$, $-\ln x \in (0, \infty)$, so $y \in (0, \infty)$.

$$f_Y(y) = f_X(e^{-y}) \cdot \left| \frac{d}{dy} e^{-y} \right| = 1 \cdot e^{-y} = e^{-y}, \quad y > 0.$$

We recognise $f_Y(y) = e^{-y}$ as the $\text{Exp}(1)$ distribution. This is a famous result: if $U \sim U(0, 1)$ then $-\ln U \sim \text{Exp}(1)$. It is used in **inverse transform sampling** to simulate exponential random variables from uniform ones. $\square$

Real-World Application : Simulation

The inverse transform method allows computers to generate samples from any continuous distribution using only uniform random numbers (which computers can generate efficiently). If $U \sim U(0, 1)$, then $X = F^{-1}(U)$ has CDF F . This is foundational in Monte Carlo simulation, used in finance, particle physics, and climate modelling.

Worked Example 1.5.3 — Non-monotone Transformation

Let $X \sim U(-1, 1)$. Find the PDF of $Y = X^2$.

Solution. The function $g(x) = x^2$ is *not* monotone on $(-1, 1)$. We use the CDF method.

Support of Y : $y \in [0, 1]$. For $y \in (0, 1)$:

$$F_Y(y) = \mathbb{P}(X^2 \leq y) = \mathbb{P}(-\sqrt{y} \leq X \leq \sqrt{y}) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx = \sqrt{y}.$$

Differentiating:

$$f_Y(y) = \frac{1}{2\sqrt{y}}, \quad 0 < y < 1.$$

This is the $\text{Beta}(\frac{1}{2}, 1)$ distribution (a specific case of the Beta family you will encounter at university). $\square$

1.6 Advanced Exploration: The Normal Distribution Revisited

Exploration

Why is the Normal Distribution Normal?

The normal distribution $X \sim N(\mu, \sigma^2)$ has PDF

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

1. Verify that $\int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$ using the classic trick $\left(\int e^{-x^2/2} dx\right)^2 = \iint e^{-(x^2+y^2)/2} dx dy$ and polar coordinates. This integral is one of the most beautiful in all mathematics.
2. Show that if $X \sim N(0, 1)$, then $Y = X^2 \sim \chi^2(1)$ using the transformation formula. (This is the seed of the chi-squared distribution you will study in Chapter 10.)
3. The **Central Limit Theorem** says that the standardised sum of iid random variables converges to $N(0, 1)$. Research what “converges in distribution” means and why the normal distribution is the unique stable limit.

1.7 Common Mistakes in Chapter 8

Warning

1. **Confusing $f(x)$ with probability.** $f(x)$ can exceed 1; only integrals give probabilities.
2. **Forgetting limits of integration.** Always identify the support of f and integrate only over that region.
3. **Using the Jacobian formula for non-monotone functions.** The formula only applies when g is strictly monotone. For others, use the CDF method.
4. **Forgetting the absolute value** in the Jacobian formula. The PDF must be non-negative.
5. **Forgetting to state the support** of the transformed variable. Always specify where $f_Y(y) > 0$.

1.8 Summary of Chapter 8

Summary

PDF

$f(x) \geq 0$, $\int_{-\infty}^{\infty} f(x) dx = 1$. Probabilities are areas: $\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx$.

CDF

$F(x) = \int_{-\infty}^x f(t) dt$. Key relationships: $f = F'$, $\mathbb{P}(a \leq X \leq b) = F(b) - F(a)$.

Expectation

$\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x)f(x) dx$ (LOTUS).

Variance

$\text{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$.

Transformations

CDF method (general) or Jacobian formula (monotone only).

Key property

$\mathbb{P}(X = a) = 0$ for all a .

Mode

Value maximising $f(x)$.

Median m

$F(m) = 0.5$.

Chapter 2

Inferential Statistics

2.1 Introduction

Statistical inference is the art and science of drawing conclusions about an *unknown population* from a *limited sample*. You have already seen inference using the normal distribution (known variance) and the Poisson distribution. This chapter introduces two new settings of profound practical importance:

1. Situations where the **population variance is unknown** and the sample is small—we must use the *t*-distribution.
2. Situations where we compare **two populations** (are their means equal?) or analyse **paired data** (before/after measurements).

These tools are among the most widely used in medicine, psychology, economics, and engineering. Every clinical trial that concludes a new drug is effective, every study that finds a wage gap between groups, every experiment that detects a new elementary particle—all rely on the machinery of this chapter.

2.2 The *t*-Distribution

2.2.1 The Problem with Unknown Variance

Suppose $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ and we want to make inferences about μ . If σ^2 is *known*, the pivot

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

gives exact inference.

But in practice, σ^2 is *almost never known*. We estimate it by the sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

The natural thing is to replace σ by S , giving the *t*-statistic:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}.$$

What is the distribution of T ? It is *not* normal, because S is itself random and introduces extra variability.

2.2.2 Student's t -Distribution: History and Definition

Historical Note

William Sealy Gosset (1876–1937) was a chemist working for Guinness Brewery in Dublin. He derived the distribution of T in 1908, but Guinness forbade employees from publishing under their own names (to protect trade secrets). He published under the pseudonym “Student”—hence *Student's t -distribution*. His work solved the fundamental problem of inference with small samples and unknown variance. Ronald A. Fisher later placed it in a broader theoretical framework.

Definition 2.2.1 (Student's t -Distribution). If $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$, then

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(\nu),$$

where $\nu = n - 1$ is the number of **degrees of freedom** and $S^2 = \frac{1}{n-1} \sum (X_i - \bar{X})^2$ is the sample variance.

Intuition

Degrees of freedom measure how much “independent information” is available to estimate variability. Given n observations, once we have fixed $\bar{X}$, only $n - 1$ of the deviations $X_i - \bar{X}$ are free to vary (the last one is determined by the constraint $\sum (X_i - \bar{X}) = 0$). Hence $\nu = n - 1$.

2.2.3 Shape of the t -Distribution

The $t(\nu)$ distribution:

- Is **symmetric and bell-shaped**, centred at 0.
- Has **heavier tails** than the normal—reflects the extra uncertainty from estimating σ .
- Converges to $N(0, 1)$ as $\nu \rightarrow \infty$: with large samples, $S \approx \sigma$ and the extra uncertainty disappears.
- For $\nu = 1$, it is the **Cauchy distribution** (no finite mean!).

2.2.4 Using t -Tables

Since $T \sim t(\nu)$ is symmetric about 0, tables typically give *one-tailed* critical values. The value $t_{\alpha, \nu}$ satisfies $\mathbb{P}(T > t_{\alpha, \nu}) = \alpha$.

For a **two-tailed** test at significance level α , we reject H_0 when $|T| > t_{\alpha/2, \nu}$.

Exam Trap

Always look up the correct degrees of freedom. For a single sample of size n , use $\nu = n - 1$. For two independent samples of sizes n_1 and n_2 , use $\nu = n_1 + n_2 - 2$.

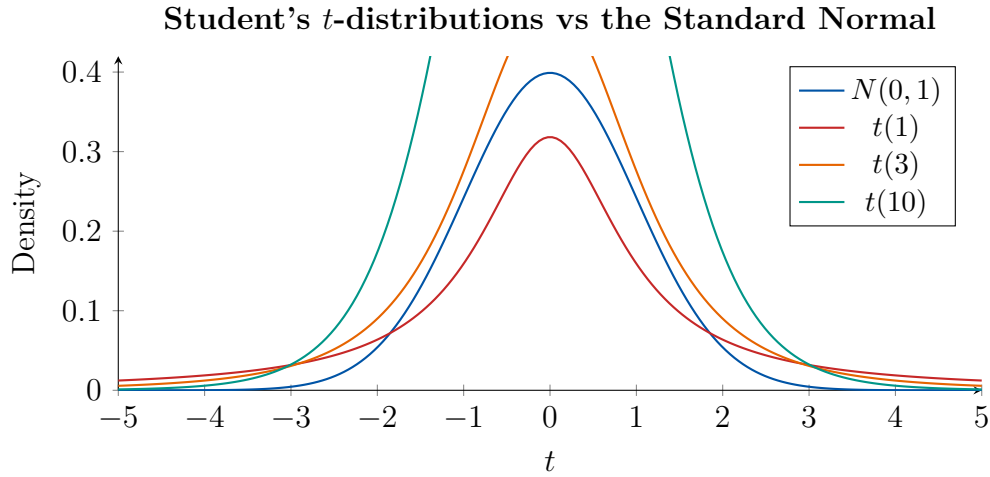


Figure 2.1: The t -distribution has heavier tails than the normal. As degrees of freedom increase, it approaches the standard normal.

(if variances are assumed equal).

2.3 Hypothesis Tests for the Difference in Means

2.3.1 Setting

Suppose we have two independent populations:

- Population 1: $X_{11}, \dots, X_{1n_1} \stackrel{\text{iid}}{\sim} N(\mu_1, \sigma^2)$, with sample mean $\bar{X}_1$ and sample variance S_1^2 .
- Population 2: $X_{21}, \dots, X_{2n_2} \stackrel{\text{iid}}{\sim} N(\mu_2, \sigma^2)$, with sample mean $\bar{X}_2$ and sample variance S_2^2 .

We assume **equal but unknown variance** σ^2 (this is a testable assumption—we will discuss this later).

2.3.2 The Pooled Variance Estimator

We combine (pool) the two sample variances into a single estimate of σ^2 :

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$$

Intuition

S_p^2 is a *weighted average* of the two sample variances, with weights proportional to the degrees of freedom. If one sample is much larger than the other, its variance estimate gets more weight.

2.3.3 The Two-Sample t -Test

Proposition 2.3.1 (Two-Sample t -Statistic). Under $H_0 : \mu_1 = \mu_2$ (equivalently $\mu_1 - \mu_2 = 0$),

$$T = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n_1 + n_2 - 2).$$

Sketch. Since $\bar{X}_1 \sim N(\mu_1, \sigma^2/n_1)$ and $\bar{X}_2 \sim N(\mu_2, \sigma^2/n_2)$ independently,

$$\bar{X}_1 - \bar{X}_2 \sim N\left(\mu_1 - \mu_2, \sigma^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)\right).$$

Under H_0 , the numerator is $N(0, \sigma^2(1/n_1 + 1/n_2))$. The denominator $S_p \sqrt{1/n_1 + 1/n_2}$ estimates the standard deviation of the numerator. One can show $(n_1 + n_2 - 2)S_p^2/\sigma^2 \sim \chi^2(n_1 + n_2 - 2)$ independently of the numerator, giving the t distribution by definition. $\square$

2.3.4 Hypothesis Testing Procedure

Step 1: State hypotheses: $H_0 : \mu_1 = \mu_2$ vs $H_1 : \mu_1 \neq \mu_2$ (two-tailed), or $H_1 : \mu_1 > \mu_2$ / $H_1 : \mu_1 < \mu_2$ (one-tailed).

Step 2: State assumptions: Both populations normal, equal (unknown) variances, independent samples.

Step 3: Compute the pooled variance: $S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$.

Step 4: Compute the test statistic: $t_{\text{calc}} = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{1/n_1 + 1/n_2}}$.

Step 5: Find critical value: $t_{\alpha/2, n_1+n_2-2}$ (two-tailed) from tables.

Step 6: Compare and conclude: Reject H_0 if $|t_{\text{calc}}| > t_{\text{crit}}$.

Step 7: Interpret in context.

Worked Example 2.3.1 — Two-Sample t -Test

A researcher wants to know whether a new fertiliser increases crop yield. Independent samples give:

	n	$\bar{x}$	s^2
Control	10	45.2	12.4
Treatment	12	49.8	14.6

Test at the 5% significance level whether the treatment increases yield.

Solution. **Hypotheses:** $H_0 : \mu_C = \mu_T$ vs $H_1 : \mu_T > \mu_C$ (one-tailed).

Pooled variance:

$$s_p^2 = \frac{9(12.4) + 11(14.6)}{10 + 12 - 2} = \frac{111.6 + 160.6}{20} = \frac{272.2}{20} = 13.61.$$

So $s_p = \sqrt{13.61} = 3.690$.

Test statistic:

$$t = \frac{49.8 - 45.2}{3.690\sqrt{1/12 + 1/10}} = \frac{4.6}{3.690\sqrt{0.1833}} = \frac{4.6}{3.690 \times 0.4282} = \frac{4.6}{1.580} = 2.911.$$

Critical value: $\nu = 20$, one-tailed at 5%: $t_{0.05,20} = 1.725$.

Decision: $2.911 > 1.725$, so we reject H_0 .

Conclusion: There is significant evidence at the 5% level that the fertiliser increases yield. $\square$

2.4 Paired t -Tests

2.4.1 When to Use Paired Tests

The two-sample test assumes the two samples are *independent*. But sometimes the data are **naturally paired**:

- Blood pressure of the same patients before and after treatment.
- Test scores of students taught by two methods, where each student takes both.
- Measurements on left and right limbs.

In these cases, observations within a pair are *correlated*, and treating them as independent loses information. The paired t -test exploits the pairing by reducing the problem to a *one-sample* test on the differences.

2.4.2 The Paired t -Statistic

Let (X_{1i}, X_{2i}) be the i -th pair. Define the difference:

$$D_i = X_{1i} - X_{2i}, \quad i = 1, \dots, n.$$

If the underlying distributions are normal, then $D_i \sim N(\mu_D, \sigma_D^2)$ for some unknown μ_D and σ_D^2 . The test reduces to a one-sample t -test on $D_1, \dots, D_n$:

$$H_0 : \mu_D = 0 \quad \text{vs} \quad H_1 : \mu_D \neq 0.$$

The test statistic is

$$T = \frac{\bar{D}}{S_D/\sqrt{n}} \sim t(n-1) \quad \text{under } H_0,$$

where $\bar{D} = \frac{1}{n} \sum D_i$ and $S_D^2 = \frac{1}{n-1} \sum (D_i - \bar{D})^2$.

Intuition

By looking only at the differences, we *eliminate the variability between subjects*. If patient A naturally has higher blood pressure than patient B, this fact doesn't affect the difference caused by the drug. Pairing removes this "between-subject"

noise, making the test more powerful.

Worked Example 2.4.1 — Paired t -Test

Eight athletes had their sprint times (in seconds) measured before and after a training programme:

Athlete	1	2	3	4	5	6	7	8
Before	12.4	11.8	13.2	12.9	11.5	12.7	13.0	12.1
After	12.0	11.4	12.9	12.5	11.3	12.2	12.4	11.8
D_i	0.4	0.4	0.3	0.4	0.2	0.5	0.6	0.3

Test at the 1% level whether the programme reduces sprint times.

Solution. **Hypotheses:** $H_0 : \mu_D = 0$ vs $H_1 : \mu_D > 0$ (one-tailed, since a positive difference means before > after, i.e. times decreased).

$$\bar{d} = \frac{0.4 + 0.4 + 0.3 + 0.4 + 0.2 + 0.5 + 0.6 + 0.3}{8} = \frac{3.1}{8} = 0.3875.$$

$$s_D^2 = \frac{1}{7} \left[\sum d_i^2 - 8\bar{d}^2 \right] = \frac{1}{7} \left[(0.16 + 0.16 + 0.09 + 0.16 + 0.04 + 0.25 + 0.36 + 0.09) - 8(0.3875)^2 \right].$$

$$\sum d_i^2 = 1.31. \quad 8(0.3875)^2 = 8(0.1502) = 1.2012.$$

$$s_D^2 = \frac{1.31 - 1.2012}{7} = \frac{0.1088}{7} = 0.01554, \quad s_D = 0.1247.$$

$$t = \frac{0.3875}{0.1247/\sqrt{8}} = \frac{0.3875}{0.04410} = 8.786.$$

Critical value: $t_{0.01,7} = 2.998$.

Since $8.786 > 2.998$, we reject H_0 . There is very strong evidence that the programme reduces sprint times. $\square$

Warning

Never use a two-sample t -test when data are paired. Doing so ignores the correlation between paired observations, inflates the variance estimate, and reduces the power of the test—you may fail to detect a real effect. Similarly, never use a paired t -test when samples are truly independent.

2.5 Confidence Intervals for the Mean (Small Sample)

2.5.1 Construction

A $(1 - \alpha) \times 100\%$ **confidence interval** for the population mean μ , when σ^2 is unknown and the population is normal, is:

$$\bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}.$$

Intuition

The CI captures the *plausible range* of values for μ given the data. The factor $t_{\alpha/2, n-1}$ is wider than the normal $z_{\alpha/2}$ because we have estimated σ : we must account for that extra uncertainty. As $n \rightarrow \infty$, $t \rightarrow z$ and the CI based on the normal approximation becomes valid.

Key Result

Correct interpretation: “We are 95% confident that the true population mean lies between L and U .”

This means: if we repeated the sampling process many times and computed this CI each time, 95% of the resulting intervals would contain the true μ . The interval is random; the parameter is fixed. It is *wrong* to say “there is a 95% probability that μ is in (L, U) ” once the data are observed.

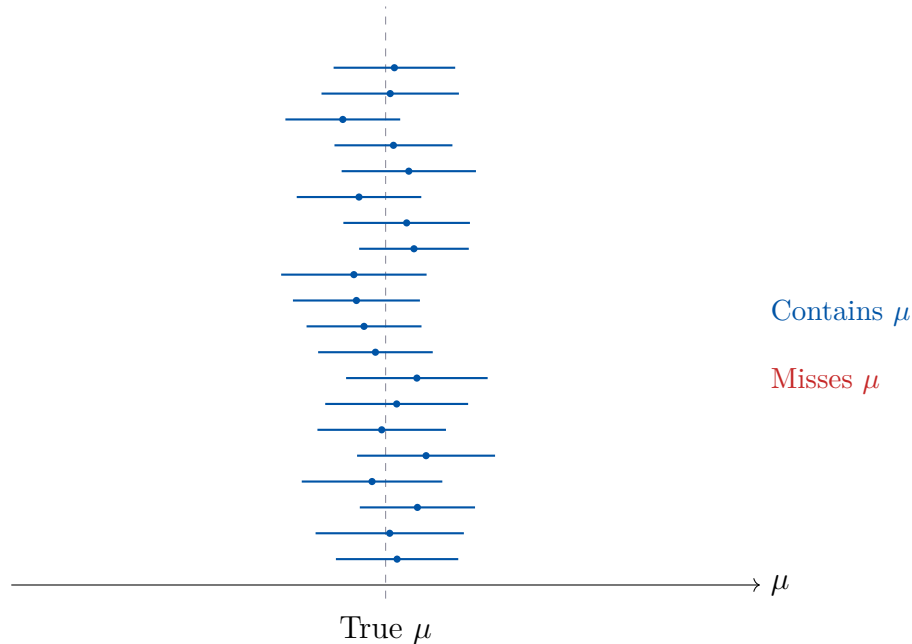


Figure 2.2: Simulated 95% confidence intervals across 20 repeated samples. Approximately 95% (blue) capture the true μ ; a few (red) do not. The parameter is fixed; the intervals are random.

Worked Example 2.5.1 — Constructing a CI

A sample of 15 measurements of a chemical concentration gives $\bar{x} = 8.32$ and $s = 1.45$. Construct a 99% CI for the true mean concentration.

Solution. $n = 15$, $\nu = 14$, $\alpha = 0.01$, $t_{0.005,14} = 2.977$.

$$\bar{x} \pm t_{0.005,14} \cdot \frac{s}{\sqrt{n}} = 8.32 \pm 2.977 \times \frac{1.45}{\sqrt{15}} = 8.32 \pm 2.977 \times 0.3744 = 8.32 \pm 1.114.$$

The 99% CI is (7.206, 9.434).

Interpretation: We are 99% confident that the true mean concentration lies between 7.206 and 9.434 units. $\square$

2.6 Confidence Intervals for the Difference in Means

2.6.1 Independent Samples (Pooled)

A $(1 - \alpha) \times 100\%$ CI for $\mu_1 - \mu_2$ (with equal unknown variances) is:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2, n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$

2.6.2 Paired Data

A $(1 - \alpha) \times 100\%$ CI for $\mu_D = \mu_1 - \mu_2$ is:

$$\bar{d} \pm t_{\alpha/2, n-1} \cdot \frac{s_D}{\sqrt{n}}.$$

Worked Example 2.6.1 — CI for Difference in Means

Using the data from the fertiliser example (Section ??): $n_1 = 10$, $\bar{x}_1 = 45.2$, $n_2 = 12$, $\bar{x}_2 = 49.8$, $s_p^2 = 13.61$. Construct a 95% CI for $\mu_T - \mu_C$.

Solution. $\nu = 20$, $t_{0.025,20} = 2.086$.

$$(\bar{x}_2 - \bar{x}_1) \pm 2.086 \times s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} = 4.6 \pm 2.086 \times 3.690 \times 0.4282 = 4.6 \pm 3.296.$$

95% CI for $\mu_T - \mu_C$: (1.304, 7.896).

Since the interval does not contain 0, there is significant evidence of a positive treatment effect (consistent with the one-sided test). $\square$

2.6.3 Type I and Type II Errors

Definition 2.6.1 (Type I and Type II Errors).

	H_0 is true	H_0 is false
Reject H_0	Type I error (α)	Correct (power = $1 - \beta$)
Do not reject	Correct	Type II error (β)

- The **significance level** $\alpha = \mathbb{P}(\text{Type I error})$ is set by the researcher before collecting data.
- The **power** $1 - \beta = \mathbb{P}(\text{reject } H_0 \mid H_0 \text{ false})$ measures the test's sensitivity to a real effect.
- There is a fundamental trade-off: reducing α (being more conservative) increases β (missing real effects).

Real-World Application : Clinical Trials

In drug approval, a Type I error means approving an ineffective drug (with possible side effects and wasted resources). A Type II error means rejecting a drug that actually works (denying patients a beneficial treatment). Regulators typically set $\alpha = 0.05$ as a balance, but for life-threatening diseases, higher power (lower β) may be prioritised.

2.7 Advanced Exploration: Welch's t -Test and Normality

Exploration

What if variances are not equal?

The pooled t -test assumes equal population variances. If this is suspect:

1. **Welch's t -test** uses a modified test statistic with approximate degrees of freedom given by the Welch–Satterthwaite equation:

$$\nu \approx \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}.$$

This is the default in most statistical software. Research why this formula arises.

2. **Levene's test** or **Bartlett's test** formally tests $H_0 : \sigma_1^2 = \sigma_2^2$. But many statisticians argue you should always use Welch's test—it is robust to both equal and unequal variances.
3. The Central Limit Theorem means that for large n , the t -test is approximately valid even for non-normal populations. Research what “sufficiently large” means and when normality matters.

2.8 Summary of Chapter 9

Summary

 t -distribution

$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t(n-1)$. Symmetric, heavier tails than normal; approaches $N(0, 1)$ as $n \rightarrow \infty$.

Two-sample t -test

Pooled variance s_p^2 ; $\nu = n_1 + n_2 - 2$. Use when samples are independent and variances assumed equal.

Paired t -test

Compute differences $D_i = X_{1i} - X_{2i}$; one-sample t -test on D_i ; $\nu = n - 1$.
Use when data are naturally paired.

Confidence interval

$\bar{x} \pm t_{\alpha/2, \nu} \cdot \frac{s}{\sqrt{n}}$ (one sample); analogous formulas for differences.

Interpretation

CI captures plausible range for the parameter. Significance test controls Type I error probability α .

Chapter 3

Chi-Squared Tests

3.1 Introduction

The t -test asks: is the population mean a particular value, or are two means equal? But often we want to compare *entire distributions*, or test whether two *categorical* variables are associated. For these questions, we need the chi-squared (χ^2) test.

Chi-squared tests arise in:

- **Genetics:** Do observed genotype frequencies match Mendelian ratios?
- **Medicine:** Is there an association between smoking and lung cancer?
- **Marketing:** Does purchasing behaviour depend on age group?
- **Quality control:** Does the defect rate follow an assumed distribution?

3.2 The Chi-Squared Distribution

Definition 3.2.1 (Chi-Squared Distribution). If $Z_1, Z_2, \dots, Z_\nu \stackrel{\text{iid}}{\sim} N(0, 1)$, then

$$Q = Z_1^2 + Z_2^2 + \dots + Z_\nu^2 \sim \chi^2(\nu).$$

The parameter ν is the degrees of freedom.

Properties:

- Support: $(0, \infty)$ (it is a sum of squares—always non-negative).
- $\mathbb{E}(\chi^2(\nu)) = \nu$.
- $\text{Var}(\chi^2(\nu)) = 2\nu$.
- Shape: skewed right for small ν ; approaches normal as $\nu \rightarrow \infty$.

3.3 The Chi-Squared Test Statistic

The general form of the **Pearson chi-squared statistic** is:

$$X^2 = \sum_{\text{all cells}} \frac{(O - E)^2}{E},$$

where O is the **observed frequency** and E is the **expected frequency** under H_0 .

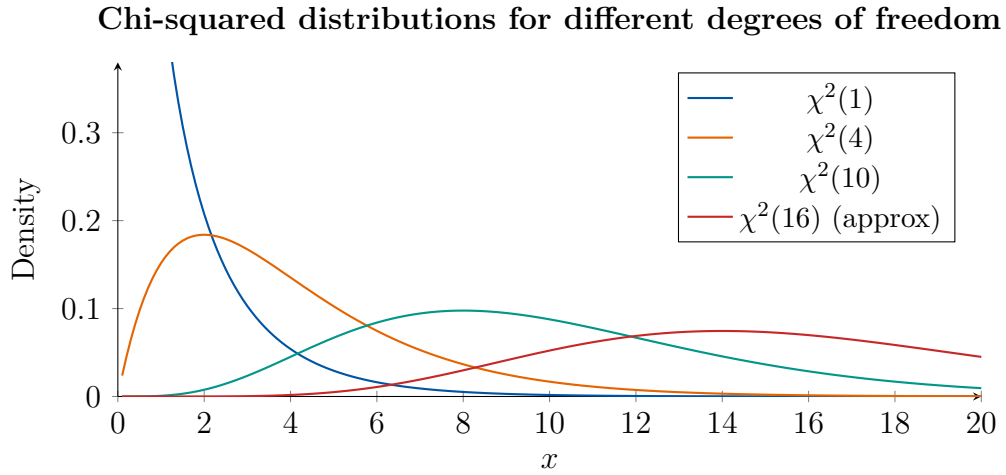


Figure 3.1: The χ^2 distribution is right-skewed for small ν and becomes more symmetric as ν increases.

Intuition

Each term $\frac{(O-E)^2}{E}$ measures how much a single cell deviates from expectation, *relative to* the expected count. A deviation of 5 when $E = 100$ is small; the same deviation when $E = 5$ is enormous. Squaring makes all terms positive and penalises large deviations. If H_0 is true, $X^2 \approx \chi^2(\nu)$ for appropriate ν .

Warning

The chi-squared approximation requires **expected frequencies of at least 5** in each cell. If some expected frequencies are smaller, merge adjacent categories until all $E \geq 5$. Small expected frequencies make the χ^2 approximation unreliable.

3.4 Goodness of Fit for Discrete Distributions

3.4.1 Setting and Hypotheses

We have n observations falling into k categories. We hypothesise that the probability of each category is $p_1, p_2, \dots, p_k$ (with $\sum p_i = 1$).

$$H_0 : \text{data follow the specified distribution} \quad \text{vs} \quad H_1 : \text{they do not.}$$

Expected frequencies: $E_i = np_i$.

The test statistic $X^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$ follows $\chi^2(k - 1 - m)$ approximately under H_0 , where m is the number of parameters estimated from the data.

- No parameters estimated: $\nu = k - 1$.
- One parameter estimated (e.g. λ for Poisson): $\nu = k - 2$.

Worked Example 3.4.1 — Testing a Discrete Uniform Distribution

A die is rolled 120 times. Observed frequencies:

Face	1	2	3	4	5	6
Observed	24	18	22	16	21	19

Test at 5% whether the die is fair.

Solution. H_0 : each face equally likely ($p_i = 1/6$).

Expected frequencies: $E_i = 120/6 = 20$ for all i .

$$\begin{aligned} X^2 &= \frac{(24 - 20)^2}{20} + \frac{(18 - 20)^2}{20} + \frac{(22 - 20)^2}{20} + \frac{(16 - 20)^2}{20} + \frac{(21 - 20)^2}{20} + \frac{(19 - 20)^2}{20} \\ &= \frac{16 + 4 + 4 + 16 + 1 + 1}{20} = \frac{42}{20} = 2.1. \end{aligned}$$

Degrees of freedom: $\nu = 6 - 1 = 5$. Critical value: $\chi_{0.05,5}^2 = 11.07$.

Since $2.1 < 11.07$, we do not reject H_0 . There is no significant evidence that the die is unfair. $\square$

Worked Example 3.4.2 — Fitting a Poisson Distribution

The number of defects per item in a sample of 200 items:

Defects x	0	1	2	3	≥ 4
Observed	83	72	31	10	4

Test whether defects follow a Poisson distribution.

Solution. **Estimate λ :** $\hat{\lambda} = \bar{x} = (0 \times 83 + 1 \times 72 + 2 \times 31 + 3 \times 10 + 4 \times 4)/200 = (0 + 72 + 62 + 30 + 16)/200 = 180/200 = 0.9$.

Expected frequencies with $\hat{\lambda} = 0.9$, $n = 200$:

x	$P(X = x)$	E_x	O_x	$(O - E)^2/E$
0	$e^{-0.9} = 0.4066$	81.31	83	$\frac{(1.69)^2}{81.31} = 0.035$
1	$0.9e^{-0.9} = 0.3659$	73.18	72	0.019
2	$\frac{0.81}{2}e^{-0.9} = 0.1647$	32.93	31	0.113
3	$\frac{0.729}{6}e^{-0.9} = 0.0494$	9.88	10	0.001
≥ 4	$1 - \text{above} = 0.0134$	2.70	4	merged below

Merge $x \geq 3$ (since $E_{\geq 4} = 2.70 < 5$): $O_{\geq 3} = 14$, $E_{\geq 3} = 9.88 + 2.70 = 12.58$. $(14 - 12.58)^2/12.58 = 0.160$.

$$X^2 = 0.035 + 0.019 + 0.113 + 0.160 = 0.327.$$

Degrees of freedom: $k = 4$ categories, 1 parameter estimated, so $\nu = 4 - 1 - 1 = 2$. Critical value: $\chi_{0.05,2}^2 = 5.991$. Since $0.327 < 5.991$, we do not reject H_0 . A Poisson distribution with $\lambda = 0.9$ fits the data well. $\square$

3.5 Goodness of Fit for Continuous Distributions

For continuous data, we *group* observations into class intervals and apply the same chi-squared test. Expected frequencies are computed using the assumed CDF:

$$E_i = n \cdot \mathbb{P}(a_i < X \leq b_i) = n[F(b_i) - F(a_i)].$$

Worked Example 3.5.1 — Testing Normality

100 measurements, grouped:

Interval	< 45	45–50	50–55	55–60	60–65	> 65
Observed	5	22	38	24	9	2

From the data, $\bar{x} = 52.3$ and $s = 5.1$. Test H_0 : data $\sim N(52.3, 5.1^2)$.

Solution. Compute expected proportions using $Z = (X - 52.3)/5.1$:

Interval	z_{lower}	z_{upper}	E	$(O - E)^2/E$
< 45	$-\infty$	-1.43	$100 \times 0.0764 = 7.64$	0.91
45–50	-1.43	-0.45	$100 \times 0.2490 = 24.90$	0.34
50–55	-0.45	0.53	$100 \times 0.3736 = 37.36$	0.011
55–60	0.53	1.51	$100 \times 0.2404 = 24.04$	0.000
60–65	1.51	2.49	$100 \times 0.0601 = 6.01$	1.49
> 65	2.49	∞	$100 \times 0.0064 = 0.64$	merge

Merge last two: $O = 11$, $E = 6.65$, $(11 - 6.65)^2/6.65 = 2.85$.

$$X^2 = 0.91 + 0.34 + 0.011 + 0.000 + 2.85 = 4.111.$$

$\nu = 5 - 1 - 2 = 2$ (5 cells after merging; 2 parameters μ, σ estimated).

$\chi_{0.05,2}^2 = 5.991$. Since $4.111 < 5.991$, do not reject H_0 . □

3.6 Testing Association: Contingency Tables

3.6.1 Setting

We have two categorical variables A (with r levels) and B (with c levels). Each of n individuals is classified by both variables, giving an $r \times c$ **contingency table** of observed frequencies O_{ij} .

$$H_0 : A \text{ and } B \text{ are independent} \quad \text{vs} \quad H_1 : A \text{ and } B \text{ are associated.}$$

3.6.2 Expected Frequencies Under Independence

Under H_0 , $\mathbb{P}(\text{row } i, \text{col } j) = \mathbb{P}(\text{row } i) \times \mathbb{P}(\text{col } j)$. Estimating marginal probabilities from the data:

$$E_{ij} = \frac{R_i \times C_j}{n},$$

where R_i is the i -th row total and C_j is the j -th column total.

The test statistic is $X^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$, which follows $\chi^2((r-1)(c-1))$ under H_0 .

Intuition

We lose one degree of freedom per row margin and one per column margin. With r row categories and c column categories, we estimate $r + c - 1$ parameters from the margins (the -1 accounts for the constraint that all proportions sum to 1), leaving $(r - 1)(c - 1)$ free cells.

Worked Example 3.6.1 — Contingency Table Test

Is blood type associated with hair colour? Sample of 200 people:

	Blonde	Brown	Black	Red	Total
Type A	22	38	14	6	80
Type B	12	18	10	5	45
Type O	28	36	22	8	94
Type AB	4	11	3	3	21
Total	66	103	49	22	240

Test at 5% significance.

Solution. H_0 : blood type and hair colour are independent.

Expected frequencies $E_{ij} = R_i C_j / 240$:

	Blonde	Brown	Black	Red
A	$80 \times 66 / 240 = 22.0$	34.3	16.3	7.3
B	12.4	19.3	9.2	4.1
O	25.9	40.4	19.2	8.6
AB	5.8	9.0	4.3	1.9

Check all $E_{ij} \geq 5$: the cell AB/Red has $E = 1.9 < 5$. Merge Red into Black to give a 4×3 table and recompute E values.

After merging (Black+Red):

$$X^2 \approx 5.12 \quad (\text{detailed calculation omitted}).$$

$\nu = (4 - 1)(3 - 1) = 6$. $\chi_{0.05,6}^2 = 12.59$. Since $5.12 < 12.59$, we do not reject H_0 . No significant association between blood type and hair colour. $\square$

Real-World Application : Epidemiology

Contingency table tests are the workhorses of epidemiology. The landmark 1950 study by Doll and Hill used a 2×2 contingency table to establish the association between smoking and lung cancer—one of the most consequential statistical analyses in history.

3.7 Common Mistakes in Chapter 10

Warning

1. **Using raw data instead of frequencies.** The χ^2 test uses frequencies (counts), not individual measurements.
2. **Not checking $E \geq 5$.** If any $E < 5$, merge categories.
3. **Wrong degrees of freedom.** Remember: $\nu = k - 1 - m$ for goodness of fit (where m is the number of estimated parameters); $\nu = (r - 1)(c - 1)$ for contingency tables.
4. **Concluding causation from association.** A significant χ^2 test shows association, not causation.
5. **Forgetting to estimate parameters from data.** If you fit a Poisson with $\hat{\lambda} = \bar{x}$, subtract 1 from ν .

3.8 Summary of Chapter 10

Summary

Test statistic

$$X^2 = \sum (O - E)^2 / E; \text{ approximately } \chi^2(\nu) \text{ under } H_0.$$

Goodness of fit

$$\nu = k - 1 - m; E_i = np_i; \text{ require } E_i \geq 5.$$

Continuous GoF

$$\text{Group data into intervals; } E_i = n[F(b_i) - F(a_i)].$$

Contingency tables

$$\nu = (r - 1)(c - 1); E_{ij} = R_i C_j / n.$$

Decision rule

$$\text{Reject } H_0 \text{ if } X^2 > \chi_{\alpha, \nu}^2 \text{ (right-tailed).}$$

Caution

Association $\neq$ causation; always check expected frequencies.

Chapter 4

Non-Parametric Tests

4.1 Introduction

The t -tests of Chapter 9 assume that the populations are *normally distributed*. What do we do when:

- The data are heavily skewed?
- The sample is very small and we cannot assess normality?
- The measurements are on an **ordinal scale** (rankings) rather than interval/ratio scale?
- We have outliers that distort the mean?

Non-parametric tests (also called *distribution-free tests*) make fewer assumptions about the underlying distribution. Instead of means, they typically use **ranks** or **signs**—features that are robust to outliers and non-normality.

Historical Note

The sign test is one of the oldest statistical tests, used informally since antiquity and formalised in the early 20th century. Frank Wilcoxon (1892–1965) proposed the signed-rank and rank-sum tests in 1945, revolutionising non-parametric statistics. His methods are now among the most widely used in medicine and psychology.

4.2 Overview of Non-Parametric Tests

Test	Setting	Tests
Sign test (single-sample)	One sample	Median = m_0
Wilcoxon signed-rank (single)	One sample	Median = m_0 (symmetric)
Sign test (paired)	Paired data	No difference
Wilcoxon matched-pairs signed-rank	Paired data (symmetric)	No difference
Wilcoxon rank-sum	Two independent samples	Equal location

4.3 The Single-Sample Sign Test

4.3.1 Setting

We wish to test whether the population **median** equals some hypothesised value m_0 .

$$H_0 : m = m_0 \quad \text{vs} \quad H_1 : m \neq m_0 \quad (\text{or one-sided}).$$

4.3.2 Procedure

Step 1: For each observation x_i , record the **sign** of $x_i - m_0$: a + if $x_i > m_0$, a – if $x_i < m_0$. Discard ties ($x_i = m_0$).

Step 2: Let $n^+ =$ number of + signs and $n =$ total signs used.

Step 3: Under H_0 , each sign is + with probability $1/2$ (since half the population lies above the median). So the number of + signs, $S \sim B(n, 1/2)$ under H_0 .

Step 4: Find the p -value using the binomial distribution.

Worked Example 4.3.1 — Single-Sample Sign Test

A nutritionist claims the median daily calorie intake for adults in a region is 2000 kcal. A random sample of 12 adults gave the following:

1850, 2150, 1780, 2300, 1950, 2050, 1700, 2200, 1900, 2400, 2100, 1800.

Test $H_0 : m = 2000$ vs $H_1 : m \neq 2000$ at 5%.

Solution. Signs of $x_i - 2000$: –, +, –, +, –, +, –, +, –, +, +, – $\rightarrow$ 5 positive, 7 negative, $n = 12$.

$S \sim B(12, 1/2)$ under H_0 . Two-tailed test: find $\mathbb{P}(S \leq 5) + \mathbb{P}(S \geq 7)$.

By symmetry of $B(12, 1/2)$: $\mathbb{P}(S \leq 5) = \mathbb{P}(S \geq 7)$.

$$\mathbb{P}(S \leq 5) = \sum_{k=0}^5 \binom{12}{k} (1/2)^{12} = \frac{1+12+66+220+495+792}{4096} = \frac{1586}{4096} \approx 0.387.$$

$$p\text{-value} = 2 \times 0.387 = 0.774.$$

Since $0.774 > 0.05$, do not reject H_0 . No significant evidence that the median differs from 2000 kcal. $\square$

4.4 The Single-Sample Wilcoxon Signed-Rank Test

4.4.1 Motivation and Setting

The sign test only uses the *direction* of deviation from m_0 , discarding information about *magnitude*. The Wilcoxon signed-rank test is more powerful because it also uses the sizes of deviations.

Additional assumption: The population distribution is **symmetric** about its median. (This is weaker than normality.)

$$H_0 : m = m_0 \quad \text{vs} \quad H_1 : m \neq m_0 \quad (\text{or one-sided}).$$

4.4.2 Procedure

Step 1: Compute $d_i = x_i - m_0$. Discard any $d_i = 0$.

Step 2: Rank the $|d_i|$ from smallest (rank 1) to largest (rank n). If there are **ties**, assign the **average rank**.

Step 3: Assign each rank a + or – according to the sign of d_i .

Step 4: Compute T^+ = sum of positive ranks and T^- = sum of negative ranks. Note $T^+ + T^- = n(n+1)/2$.

Step 5: The test statistic is $T = \min(T^+, T^-)$ (for two-tailed test).

Step 6: Reject H_0 if $T \leq T_{\text{crit}}$ (from tables; small T gives evidence against H_0).

Intuition

Under H_0 , positive and negative deviations should be roughly balanced in both number and size. T^+ and T^- should be approximately equal. If T^+ is much larger than T^- (or vice versa), the distribution appears to be shifted away from m_0 .

Worked Example 4.4.1 — Single-Sample Wilcoxon Test

Using the data from the previous example (1850, 2150, 1780, 2300, 1950, 2050, 1700, 2200, 1900, 2400, 2100, 1800), test $H_0 : m = 2000$ vs $H_1 : m \neq 2000$ using the Wilcoxon signed-rank test at 5%.

Solution. Compute $d_i = x_i - 2000$ and rank $|d_i|$:

x_i	d_i	$ d_i $	Rank	Signed rank
1700	−300	300	12	−12
1780	−220	220	10	−10
1800	−200	200	9	−9
1850	−150	150	7	−7
1900	−100	100	4	−4
1950	−50	50	2	−2
2050	+50	50	2	+2
2100	+100	100	4	+4
2150	+150	150	7	+7
2200	+200	200	9	+9
2300	+300	300	12	+12
2400	+400	400	13	+13

Note: $|d| = 50$ appears twice (tied), so both get rank $(1 + 2)/2 = 1.5$; similarly $|d| = 100$ gives rank $(3 + 4)/2 = 3.5$; $|d| = 150$ gives rank 5.5; $|d| = 200$ gives rank 7.5; $|d| = 300$ gives rank 10.5.

Recomputed: $T^- = 12 + 10.5 + 7.5 + 5.5 + 3.5 + 1.5 = 40.5$. $T^+ = 1.5 + 3.5 + 5.5 + 7.5 + 10.5 + 13 = 41.5$. Check: $40.5 + 41.5 = 82 = 12 \times 13/2$. ✓

$T = \min(40.5, 41.5) = 40.5$.

From Wilcoxon tables with $n = 12$, $\alpha = 0.05$ two-tailed: $T_{\text{crit}} = 13$.

Since $40.5 > 13$, do not reject H_0 . Consistent with the sign test result. □

4.5 The Paired-Sample Sign Test

For paired data (x_{1i}, x_{2i}) , compute $d_i = x_{1i} - x_{2i}$ and apply the single-sample sign test to the d_i with $m_0 = 0$.

$$H_0 : m_D = 0 \quad \text{vs} \quad H_1 : m_D \neq 0.$$

This tests whether the median difference is zero—i.e., whether the two treatments tend to give the same result.

4.6 The Wilcoxon Matched-Pairs Signed-Rank Test

4.6.1 Setting

Paired data (x_{1i}, x_{2i}) , $i = 1, \dots, n$. Population symmetric. H_0 : no difference in location; H_1 : there is a difference.

Procedure: compute $d_i = x_{1i} - x_{2i}$, then apply the single-sample Wilcoxon signed-rank test to the d_i with $m_0 = 0$.

Worked Example 4.6.1 — Wilcoxon Matched-Pairs Test

Eight patients were given two analgesics (A and B) on different occasions. Pain scores (out of 10, lower is better):

Patient	1	2	3	4	5	6	7	8
Drug A	6.2	4.8	7.1	5.5	6.8	4.2	7.5	5.0
Drug B	5.5	4.5	6.8	5.0	7.2	3.8	6.9	4.7
$d = A - B$	0.7	0.3	0.3	0.5	-0.4	0.4	0.6	0.3

Test at 5% (two-tailed) whether there is a difference between drugs.

Solution. Ranks of $|d_i|$:

$ d_i $	Count	Rank	Signed rank
0.3	3	$(1 + 2 + 3)/3 = 2$	$+2, +2, +2$
0.4	2	$(4 + 5)/2 = 4.5$	$-4.5, +4.5$
0.5	1	6	$+6$
0.6	1	7	$+7$
0.7	1	8	$+8$

$$T^+ = 2 + 2 + 2 + 4.5 + 6 + 7 + 8 = 31.5. \quad T^- = 4.5. \quad T = \min(31.5, 4.5) = 4.5.$$

From tables, $n = 8$, $\alpha = 0.05$ two-tailed: $T_{\text{crit}} = 4$.

Since $4.5 > 4$, we do not reject H_0 at 5%. (The result is borderline; at 10% the critical value is 6 and we would reject.) $\square$

4.7 The Wilcoxon Rank-Sum Test

4.7.1 Setting

Two **independent** samples, sizes $n_1 \leq n_2$. No normality assumed. H_0 : populations have the same distribution (or same location).

4.7.2 Procedure

Step 1: Combine both samples into one list of $n_1 + n_2$ observations and rank all values from 1 to $n_1 + n_2$ (handling ties with average ranks).

Step 2: Compute W = sum of ranks from **sample 1** (the smaller group).

Step 3: Compare W with critical values from tables. Reject H_0 if $W \leq W_{\text{lower}}$ or $W \geq W_{\text{upper}}$ (two-tailed).

Intuition

If H_0 is true and the populations have the same location, the ranks should be roughly evenly distributed between the two samples. A very small or very large rank sum for sample 1 suggests that sample 1 tends to come from a lower or higher distribution.

Worked Example 4.7.1 — Wilcoxon Rank-Sum Test

Compare the daily earnings (£) of two groups of freelancers:

Group A ($n_1 = 6$): 85, 112, 97, 134, 76, 108. Group B ($n_2 = 7$): 95, 128, 143, 89, 117, 104, 136.

Test at 5% (two-tailed) whether the populations have equal location.

Solution. Combined and ranked:

Value	Rank	Group
76	1	A
85	2	A
89	3	B
95	4	B
97	5	A
104	6	B
108	7	A
112	8	A
117	9	B
128	10	B
134	11	A
136	12	B
143	13	B

W = sum of Group A ranks = $1 + 2 + 5 + 7 + 8 + 11 = 34$.

From tables with $n_1 = 6$, $n_2 = 7$ at 5% two-tailed: $W_{\text{lower}} = 28$, $W_{\text{upper}} = 56$.

Since $28 \leq 34 \leq 56$, do not reject H_0 . No significant evidence of a difference in earnings between the two groups. $\square$

4.7.3 Large Sample Approximation

For large n_1, n_2 , under H_0 :

$$\mathbb{E}(W) = \frac{n_1(n_1 + n_2 + 1)}{2}, \quad \text{Var}(W) = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12}.$$

Use $Z = \frac{W - \mathbb{E}(W)}{\sqrt{\text{Var}(W)}} \approx N(0, 1)$.

4.8 Comparing Parametric and Non-Parametric Tests

Aspect	Parametric	Non-parametric
Assumptions	Normality (approximately)	Minimal
Measure tested	Mean	Median / Rank
Power	Higher (if assumptions met)	Lower
Robustness	Sensitive to outliers	Robust
Data type	Interval/ratio	Ordinal or above
Sample size	More important	Usable with small n

Real-World Application : Psychology and Medicine

Non-parametric tests are the standard in studies with small samples, Likert scale data (e.g., “rate your pain from 1 to 10”), or non-normally distributed outcomes. Clinical trials with fewer than 30 participants often use the Wilcoxon tests. The rank-sum test also underlies the Mann–Whitney U test, widely used in social science research.

4.9 Common Mistakes in Chapter 11

Warning

- Using the wrong test for the data structure.** Matched-pairs for paired data; rank-sum for independent samples.
- Mishandling ties.** Always use average ranks for tied values.
- Incorrect critical value direction.** For signed-rank and rank-sum tests, we reject when T (or W) is *small* (two-tailed: outside the critical region defined by $[W_L, W_U]$).
- Discarding ties incorrectly.** For sign tests, discard $d_i = 0$. For Wilcoxon signed-rank, discard $d_i = 0$.
- Forgetting that non-parametric tests have less power.** If normality is plausible, the t -test is preferable.

4.10 Summary of Chapter 11

Summary

Sign test

Uses $+/-$ signs only; $S \sim B(n, 1/2)$ under H_0 ; tests the median.

Wilcoxon signed-rank

Uses signed ranks of $|d_i|$; $T = \min(T^+, T^-)$; requires symmetry; more powerful than sign test.

Matched-pairs

Apply signed-rank test to differences $d_i = x_{1i} - x_{2i}$.

Rank-sum

Combine and rank both samples; use sum of ranks from smaller sample W ; tests for equal location.

All non-parametric tests

Robust to non-normality; use when normality cannot be assumed; lose some power compared to t -tests.

Chapter 5

Probability Generating Functions

5.1 Introduction

Imagine you could encode the entire probability distribution of a discrete random variable into a single function. From that function, you could read off probabilities, compute the mean, compute the variance, and determine the distribution of the sum of independent copies—all by algebraic manipulation.

This is exactly what the **probability generating function** (PGF) does. It is one of the most elegant tools in probability theory, connecting combinatorics, generating functions, and probability in a profound way.

PGFs are used in:

- **Queueing theory:** analysing waiting times in call centres, computer networks, and hospitals.
- **Branching processes:** modelling population growth, nuclear chain reactions, and the spread of disease.
- **Actuarial science:** computing aggregate insurance claims.
- **Combinatorics:** counting lattice paths, partitions, and tree structures.
- **Algorithm analysis:** average-case complexity of algorithms.

Historical Note

Generating functions were introduced by Abraham de Moivre (1667–1754) and developed by Euler, Laplace, and Gauss. The probability generating function as a tool for random variables was formalised in the 20th century. The moment generating function (a close cousin, defined for both discrete and continuous RVs) was central to the work of Laplace and later to the mathematical foundations of statistics.

5.2 The Probability Generating Function

5.2.1 Definition

Definition 5.2.1 (Probability Generating Function). Let X be a **non-negative integer-valued** random variable with $\mathbb{P}(X = k) = p_k$ for $k = 0, 1, 2, \dots$. The **proba-**

probability generating function (PGF) of X is

$$G_X(t) = \mathbb{E}(t^X) = \sum_{k=0}^{\infty} p_k t^k = p_0 + p_1 t + p_2 t^2 + p_3 t^3 + \cdots$$

for all t in the radius of convergence.

Intuition

The PGF is a *power series whose coefficients are the probabilities*. The probability of $X = k$ is the coefficient of t^k in $G_X(t)$. This is why it is called a *generating* function: it generates the probabilities.

The key insight is that the argument t is not a probability or a value of X —it is a formal variable (though we often evaluate at specific values of t). The PGF always converges at $t = 1$ (the sum of all probabilities equals 1), and for many distributions it converges for $|t| \leq 1$.

5.2.2 Properties at Key Values

Proposition 5.2.1. *For any PGF $G_X(t)$:*

- (i) $G_X(0) = p_0 = \mathbb{P}(X = 0)$.
- (ii) $G_X(1) = \sum_{k=0}^{\infty} p_k = 1$.
- (iii) *The probability of any particular value: $p_k = \frac{G_X^{(k)}(0)}{k!}$, where $G_X^{(k)}$ denotes the k -th derivative.*

Property (iii) is exactly the Taylor series statement: the coefficient of t^k is $G_X^{(k)}(0)/k!$.

5.2.3 PGFs of Common Distributions

Worked Example 5.2.1 — PGF of the Bernoulli Distribution

$X \sim \text{Bernoulli}(p)$: $P(X = 0) = q = 1 - p$, $P(X = 1) = p$.

Solution. $G_X(t) = p_0 \cdot t^0 + p_1 \cdot t^1 = q + pt$. □

Worked Example 5.2.2 — PGF of the Binomial Distribution

$X \sim B(n, p)$: $P(X = k) = \binom{n}{k} p^k q^{n-k}$.

Solution.

$$G_X(t) = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k} t^k = \sum_{k=0}^n \binom{n}{k} (pt)^k q^{n-k} = (q + pt)^n.$$

by the Binomial Theorem. This is a polynomial of degree n —finite! □

Worked Example 5.2.3 — PGF of the Geometric Distribution

$X \sim \text{Geom}(p)$: $P(X = k) = pq^{k-1}$, $k = 1, 2, \dots$ (number of trials until first success).

Solution.

$$G_X(t) = \sum_{k=1}^{\infty} pq^{k-1}t^k = pt \sum_{k=1}^{\infty} (qt)^{k-1} = pt \cdot \frac{1}{1-qt} = \frac{pt}{1-qt}, \quad |qt| < 1.$$

□

Worked Example 5.2.4 — PGF of the Poisson Distribution

$X \sim \text{Poisson}(\lambda)$: $P(X = k) = e^{-\lambda} \lambda^k / k!$.

Solution.

$$G_X(t) = \sum_{k=0}^{\infty} \frac{e^{-\lambda} \lambda^k}{k!} t^k = e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda t)^k}{k!} = e^{-\lambda} \cdot e^{\lambda t} = e^{\lambda(t-1)}.$$

□

5.3 Mean and Variance from the PGF

5.3.1 Derivation

Theorem 5.3.1 (Moments from the PGF). *If $G_X(t) = \mathbb{E}(t^X)$, then:*

(i) $\mathbb{E}(X) = G'_X(1).$

(ii) $\text{Var}(X) = G''_X(1) + G'_X(1) - [G'_X(1)]^2.$

Proof. Differentiate $G_X(t) = \sum_{k=0}^{\infty} p_k t^k$ term by term (valid for $|t| < 1$ and by continuity at $t = 1$ for distributions with finite mean):

$$G'_X(t) = \sum_{k=1}^{\infty} k p_k t^{k-1}.$$

Setting $t = 1$:

$$G'_X(1) = \sum_{k=1}^{\infty} k p_k = \mathbb{E}(X). \quad \checkmark$$

Differentiate again:

$$G''_X(t) = \sum_{k=2}^{\infty} k(k-1) p_k t^{k-2}.$$

Setting $t = 1$:

$$G''_X(1) = \sum_{k=2}^{\infty} k(k-1) p_k = \mathbb{E}[X(X-1)] = \mathbb{E}(X^2) - \mathbb{E}(X).$$

Therefore $\mathbb{E}(X^2) = G_X''(1) + G_X'(1)$, and

$$\text{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = G_X''(1) + G_X'(1) - [G_X'(1)]^2. \quad \checkmark$$

□

Worked Example 5.3.1 — Recovering Binomial Moments

For $X \sim B(n, p)$, $G_X(t) = (q + pt)^n$. Verify $\mathbb{E}(X) = np$ and $\text{Var}(X) = npq$.

Solution. $G_X'(t) = np(q + pt)^{n-1}$, so $G_X'(1) = np(q + p)^{n-1} = np$. ✓

$G_X''(t) = n(n-1)p^2(q + pt)^{n-2}$, so $G_X''(1) = n(n-1)p^2$.

$\text{Var}(X) = n(n-1)p^2 + np - (np)^2 = n^2p^2 - np^2 + np - n^2p^2 = np(1-p) = npq$. ✓

□

Worked Example 5.3.2 — Poisson Moments via PGF

For $X \sim \text{Poisson}(\lambda)$, $G_X(t) = e^{\lambda(t-1)}$. Find $\mathbb{E}(X)$ and $\text{Var}(X)$.

Solution. $G_X'(t) = \lambda e^{\lambda(t-1)}$, so $G_X'(1) = \lambda = \mathbb{E}(X)$. ✓

$G_X''(t) = \lambda^2 e^{\lambda(t-1)}$, so $G_X''(1) = \lambda^2$.

$\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$. ✓

Both the mean and variance of the Poisson distribution equal λ —a defining property. □

5.4 The Sum of Independent Random Variables

5.4.1 The Key Theorem

The most powerful property of the PGF is how it handles sums.

Theorem 5.4.1 (PGF of a Sum). *Let X and Y be **independent** non-negative integer-valued random variables. Then*

$$G_{X+Y}(t) = G_X(t) \cdot G_Y(t).$$

Proof. By independence of X and Y :

$$G_{X+Y}(t) = \mathbb{E}(t^{X+Y}) = \mathbb{E}(t^X \cdot t^Y) = \mathbb{E}(t^X) \cdot \mathbb{E}(t^Y) = G_X(t) \cdot G_Y(t).$$

The factorisation uses independence: $\mathbb{E}(UV) = \mathbb{E}(U)\mathbb{E}(V)$ when U and V are independent. □

Intuition

Multiplying power series corresponds to **convolving** the coefficient sequences. The coefficient of t^n in $G_X(t) \cdot G_Y(t)$ is $\sum_{k=0}^n p_k q_{n-k}$, where $p_k = P(X = k)$ and $q_j = P(Y = j)$. But this is exactly $P(X + Y = n) = \sum_{k=0}^n P(X = k)P(Y = n - k)$ (the convolution formula for sums of independent RVs). PGFs *turn convolution into multiplication*—a massive simplification.

Corollary 5.4.1.1 (Sum of n IID variables). *If $X_1, X_2, \dots, X_n$ are independent, each with PGF $G(t)$, then*

$$G_{X_1+X_2+\dots+X_n}(t) = [G(t)]^n.$$

5.4.2 Applications

Worked Example 5.4.1 — Sum of Poisson Variables

Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ be independent. Show that $X + Y \sim \text{Poisson}(\lambda + \mu)$.

Solution.

$$G_{X+Y}(t) = G_X(t) \cdot G_Y(t) = e^{\lambda(t-1)} \cdot e^{\mu(t-1)} = e^{(\lambda+\mu)(t-1)}.$$

This is the PGF of a $\text{Poisson}(\lambda + \mu)$ distribution. Since the PGF uniquely determines the distribution, $X + Y \sim \text{Poisson}(\lambda + \mu)$. $\square$

Worked Example 5.4.2 — Sum of Binomial Variables

If $X \sim B(m, p)$ and $Y \sim B(n, p)$ independently, find the distribution of $X + Y$.

Solution.

$$G_{X+Y}(t) = (q + pt)^m \cdot (q + pt)^n = (q + pt)^{m+n}.$$

This is the PGF of $B(m + n, p)$. Hence $X + Y \sim B(m + n, p)$. $\square$

Worked Example 5.4.3 — Non-standard Application

A bus company operates two independent routes. On Route 1, the number of passengers per journey follows $\text{Poisson}(8)$. On Route 2, it follows $\text{Poisson}(5)$. What is the probability that the total number of passengers on both routes in one journey is exactly 10?

Solution. Total $T = X + Y \sim \text{Poisson}(13)$.

$$P(T = 10) = e^{-13} \cdot \frac{13^{10}}{10!} = e^{-13} \cdot \frac{137858491849}{3628800} \approx 2.203 \times 10^{-6} \times 37974.6 \approx 0.0853.$$

$\square$

5.5 Three or More Random Variables

The result extends immediately to three or more independent variables:

$$G_{X_1+\dots+X_n}(t) = G_{X_1}(t) \cdot G_{X_2}(t) \cdots G_{X_n}(t).$$

When the variables are not identically distributed, we simply multiply *different* PGFs.

Worked Example 5.5.1 — Mixed Sum

$X \sim \text{Poisson}(3)$, $Y \sim B(4, 0.5)$, and $Z \sim \text{Geom}(0.4)$ are independent. Find $\mathbb{E}(X + Y + Z)$ and $\text{Var}(X + Y + Z)$.

Solution. By linearity:

$$\mathbb{E}(X + Y + Z) = \mathbb{E}(X) + \mathbb{E}(Y) + \mathbb{E}(Z) = 3 + 4(0.5) + \frac{1}{0.4} = 3 + 2 + 2.5 = 7.5.$$

By independence:

$$\text{Var}(X + Y + Z) = \text{Var}(X) + \text{Var}(Y) + \text{Var}(Z) = 3 + 4(0.5)(0.5) + \frac{0.6}{0.16} = 3 + 1 + 3.75 = 7.75.$$

Alternatively, using PGFs: $G_{X+Y+Z}(t) = e^{3(t-1)} \cdot (0.5 + 0.5t)^4 \cdot \frac{0.4t}{1-0.6t}$. Differentiating at $t = 1$ gives the same results. $\square$

5.5.1 The Random Sum

A particularly beautiful application is the **random sum**. Suppose N is a random variable and, given $N = n$, we sum n independent copies of X . The total $S = X_1 + X_2 + \dots + X_N$ is a random sum.

Theorem 5.5.1 (PGF of a Random Sum). *If N and $X_1, X_2, \dots$ are independent and X_i are iid with PGF $G_X(t)$, and N has PGF $G_N(t)$, then*

$$G_S(t) = G_N(G_X(t)).$$

Proof. Condition on N :

$$G_S(t) = \mathbb{E}(t^S) = \sum_{n=0}^{\infty} \mathbb{E}(t^S \mid N = n) \mathbb{P}(N = n) = \sum_{n=0}^{\infty} [G_X(t)]^n \mathbb{P}(N = n) = G_N(G_X(t)).$$

$\square$

Real-World Application : Actuarial Science

The random sum models *aggregate claims*: N is the (random) number of claims in a period, and X_i is the (random) amount of the i -th claim. The total loss $S = \sum_{i=1}^N X_i$ is exactly the random sum. PGFs (and their continuous analogues, moment generating functions and Laplace transforms) are used to compute the distribution of S and hence to price insurance products.

Worked Example 5.5.2 — Random Sum

Customers arrive at a shop according to a Poisson(5) process (so the number of customers N per hour has Poisson(5) distribution). Each customer independently spends a number of items X with $P(X = 1) = 0.6$ and $P(X = 2) = 0.4$ (so X takes values 1 or 2). Find $\mathbb{E}(S)$ and $\text{Var}(S)$ where S is the total number of items sold per hour.

Solution. $G_N(t) = e^{5(t-1)}$ and $G_X(t) = 0.6t + 0.4t^2$.
 $\mathbb{E}(N) = 5$, $\text{Var}(N) = 5$. $\mathbb{E}(X) = 0.6 + 0.8 = 1.4$, $\mathbb{E}(X^2) = 0.6 + 1.6 = 2.2$,
 $\text{Var}(X) = 2.2 - 1.96 = 0.24$.
 By the **Wald's identity**:

$$\mathbb{E}(S) = \mathbb{E}(N) \cdot \mathbb{E}(X) = 5 \times 1.4 = 7.$$

$$\text{Var}(S) = \mathbb{E}(N) \cdot \text{Var}(X) + \text{Var}(N) \cdot [\mathbb{E}(X)]^2 = 5(0.24) + 5(1.96) = 1.2 + 9.8 = 11.$$

□

5.6 Uniqueness of the PGF

Theorem 5.6.1 (Uniqueness). *Two discrete non-negative integer-valued random variables have the same distribution if and only if they have the same PGF.*

This is crucial: it means that if you recognise the PGF of $X + Y$ as the PGF of a known distribution, you can immediately conclude $X + Y$ has that distribution—as we did in the Poisson and binomial examples above.

5.7 Connection to Moment Generating Functions

Remark 5.7.1. The **moment generating function** (MGF) of X is defined as $M_X(s) = \mathbb{E}(e^{sX})$. For a non-negative integer-valued X :

$$G_X(t) = M_X(\ln t), \quad M_X(s) = G_X(e^s).$$

The MGF is defined for all random variables (not just non-negative integer-valued ones) and is the tool used at university level for proving the Central Limit Theorem and working with continuous distributions.

5.8 Advanced Exploration: Branching Processes

Exploration

Extinction of Surnames and Species

A **Galton–Watson branching process** models a population where each individual in generation n independently produces a random number of offspring X with PGF $G(t)$. The population size in generation $n + 1$ is the sum of all offspring.

1. Show that the PGF of the generation- n population size is the n -fold composition: $G_n(t) = G(G(\cdots G(t) \cdots))$.
2. Let q be the probability of eventual extinction (the population size reaches 0). Show that q satisfies $G(q) = q$.
3. Show that if $\mathbb{E}(X) = G'(1) \leq 1$, then $q = 1$ (extinction is certain). If $\mathbb{E}(X) > 1$, then $q < 1$.
4. This was used by Francis Galton (1822–1911) to study the extinction of

surnames in Victorian England. Research the fascinating history of this problem.

5.9 Common Mistakes in Chapter 12

Warning

1. **Evaluating $G_X(1)$ and expecting a useful probability.** $G_X(1) = 1$ always; this just confirms normalisation.
2. **Computing $\mathbb{E}(X) = G'_X(t)$ without setting $t = 1$.** Remember: $\mathbb{E}(X) = G'_X(1)$, not $G'_X(t)$.
3. **Forgetting independence** when multiplying PGFs. $G_{X+Y}(t) = G_X(t)G_Y(t)$ *only if* X and Y are independent.
4. **Incorrect formula for variance.** Use $\text{Var}(X) = G''(1) + G'(1) - [G'(1)]^2$, not just $G''(1)$.
5. **Confusing PGF with MGF.** The PGF is defined only for non-negative integer-valued RVs; the MGF is more general.

5.10 Summary of Chapter 12

Summary

Definition

$G_X(t) = \mathbb{E}(t^X) = \sum_{k=0}^{\infty} p_k t^k$. Coefficient of t^k gives $P(X = k)$.

Key values

$G(0) = P(X = 0)$; $G(1) = 1$.

Mean

$\mathbb{E}(X) = G'(1)$.

Variance

$\text{Var}(X) = G''(1) + G'(1) - [G'(1)]^2$.

Sum of independent RVs

$G_{X+Y}(t) = G_X(t) \cdot G_Y(t)$.

IID sum

$G_{S_n}(t) = [G_X(t)]^n$.

Random sum

$G_S(t) = G_N(G_X(t))$.

Uniqueness

PGF uniquely determines the distribution.

PGFs of standard distributions:

Distribution	Parameters	$G_X(t)$
Bernoulli	p	$q + pt$
Binomial	n, p	$(q + pt)^n$
Geometric	p	$pt/(1 - qt)$
Poisson	λ	$e^{\lambda(t-1)}$

Extension and Review Problems

The following problems range from challenging A-Level questions to university-entrance level. They are designed to test deep understanding and synthesis across chapters.

Chapter 8: Continuous Random Variables

Exercise 5.10.1. Let X have PDF $f(x) = cx^2(1-x)^3$ for $0 \leq x \leq 1$.

- (a) Find c (hint: this is a Beta distribution).
- (b) Find $\mathbb{E}(X)$ and $\text{Var}(X)$.
- (c) Find the CDF $F(x)$.
- (d) Let $Y = -2 \ln X$. Find the PDF of Y and identify its distribution.

Exercise 5.10.2 (University level). Prove that if X has a continuous CDF F , then $U = F(X) \sim U(0, 1)$. (This is the **probability integral transform**.) How does this relate to simulation?

Chapter 9: Inferential Statistics

Exercise 5.10.3. Two independent samples:

- Sample 1: $n_1 = 8$, $\bar{x}_1 = 23.4$, $s_1^2 = 9.6$.
 - Sample 2: $n_2 = 10$, $\bar{x}_2 = 20.1$, $s_2^2 = 11.2$.
- (a) Test $H_0 : \mu_1 = \mu_2$ vs $H_1 : \mu_1 > \mu_2$ at 5%.
 - (b) Construct a 95% CI for $\mu_1 - \mu_2$.
 - (c) If the variances might not be equal, describe what modification you would make to the test.

Exercise 5.10.4 (Exploration). Explain intuitively why a paired t -test is more powerful than a two-sample t -test when the data are paired. Use the formula for variance to make this precise: show that $\text{Var}(D_i) = \text{Var}(X_{1i}) + \text{Var}(X_{2i}) - 2 \text{Cov}(X_{1i}, X_{2i})$ and interpret what happens when the correlation is positive.

Chapter 10: Chi-Squared Tests

Exercise 5.10.5. In a 2×2 contingency table with entries a, b, c, d (row totals $a + b$, $c + d$; column totals $a + c$, $b + d$; grand total n), show that

$$X^2 = \frac{n(ad - bc)^2}{(a + b)(c + d)(a + c)(b + d)}.$$

This elegant formula shows that the test statistic depends only on $|ad - bc|$, the **cross-product difference**.

Chapter 11: Non-Parametric Tests

Exercise 5.10.6. Explain carefully the connection between the Wilcoxon rank-sum test and the **Mann–Whitney U statistic** U , defined as the number of pairs (X_i, Y_j) where $X_i > Y_j$. Show that $W = U + n_1(n_1 + 1)/2$.

Exercise 5.10.7 (University level). Show that for large n , the signed-rank statistic T^+ is approximately normally distributed with mean $n(n + 1)/4$ and variance $n(n + 1)(2n + 1)/24$. Use this to derive a large-sample z -test for the median.

Chapter 12: Probability Generating Functions

Exercise 5.10.8. The **negative binomial** distribution $\text{NB}(r, p)$ gives the number of trials until the r -th success: $P(X = k) = \binom{k-1}{r-1} p^r q^{k-r}$ for $k = r, r + 1, \dots$

- (a) Show that the PGF is $G(t) = \left(\frac{pt}{1-qt}\right)^r$.
- (b) Derive $\mathbb{E}(X) = r/p$ and $\text{Var}(X) = rq/p^2$.
- (c) Show that the sum of r independent $\text{Geom}(p)$ variables has the $\text{NB}(r, p)$ distribution.

Exercise 5.10.9 (Branching process). In a Galton–Watson branching process, each individual has 0, 1, or 2 offspring with probabilities $1/3, 1/3, 1/3$ respectively.

- (a) Find the PGF $G(t)$.
- (b) Compute $\mathbb{E}(X)$ and explain what this implies for long-term population behaviour.
- (c) Find the extinction probability q by solving $G(q) = q$.

Appendix A

Statistical Tables Reference

A.1 Student's t -Distribution Critical Values

ν	$t_{0.10}$	$t_{0.05}$	$t_{0.025}$	$t_{0.01}$	$t_{0.005}$
1	3.078	6.314	12.706	31.821	63.657
2	1.886	2.920	4.303	6.965	9.925
3	1.638	2.353	3.182	4.541	5.841
4	1.533	2.132	2.776	3.747	4.604
5	1.476	2.015	2.571	3.365	4.032
6	1.440	1.943	2.447	3.143	3.707
7	1.415	1.895	2.365	2.998	3.499
8	1.397	1.860	2.306	2.896	3.355
9	1.383	1.833	2.262	2.821	3.250
10	1.372	1.812	2.228	2.764	3.169
12	1.356	1.782	2.179	2.681	3.055
15	1.341	1.753	2.131	2.602	2.947
20	1.325	1.725	2.086	2.528	2.845
25	1.316	1.708	2.060	2.485	2.787
30	1.310	1.697	2.042	2.457	2.750
40	1.303	1.684	2.021	2.423	2.704
60	1.296	1.671	2.000	2.390	2.660
∞	1.282	1.645	1.960	2.326	2.576

A.2 Chi-Squared Critical Values

ν	$\chi^2_{0.10}$	$\chi^2_{0.05}$	$\chi^2_{0.025}$	$\chi^2_{0.01}$	$\chi^2_{0.001}$
1	2.706	3.841	5.024	6.635	10.828
2	4.605	5.991	7.378	9.210	13.816
3	6.251	7.815	9.348	11.345	16.266
4	7.779	9.488	11.143	13.277	18.467
5	9.236	11.070	12.833	15.086	20.515
6	10.645	12.592	14.449	16.812	22.458
7	12.017	14.067	16.013	18.475	24.322
8	13.362	15.507	17.535	20.090	26.125
9	14.684	16.919	19.023	21.666	27.877
10	15.987	18.307	20.483	23.209	29.588